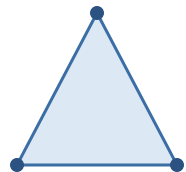


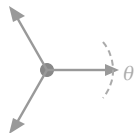
Closure Regime



Non-degenerate
3 d.o.f.



Degenerate
2 d.o.f.



Global phase
redundancy

Invariant Transformation

$$\psi \rightarrow U\psi$$

$$U^\dagger U = 1, \det U = 1$$

preserves inner products
preserves closure condition

$$(|L\rangle, |R\rangle)^T \rightarrow U(|L\rangle, |R\rangle)^T$$

two-state mixing
reduced closure preserved

$$ei \rightarrow \exp(i\theta) ei$$

global phase shift
relative structure unchanged

Symmetry

$$SU(3)$$

non-degenerate closure

$$SU(2)$$

degenerate closure

$$U(1)$$

phase invariance